Exhaustive Search for Aerodynamically Optimal Aircraft Configurations: A First-Principles Framework with Semi-Empirical Correlations

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Abstract

We present a systematic approach to identifying aerodynamically optimal subsonic aircraft configurations through exhaustive parameter space search. Our methodology combines first-principles aerodynamic equations with carefully documented semi-empirical correlations, explicitly distinguishing between physically grounded relationships and heuristic surrogate models. We evaluate over 10 million design candidates across a 5-dimensional parameter space comprising aspect ratio, sweep angle, taper ratio, thickness-to-chord ratio, and camber. Our best design achieves a lift-to-drag ratio of 27.74 at an aspect ratio of 18.3, with all solutions satisfying declared constraints.

Keywords: Aircraft design optimization, exhaustive search, aerodynamic efficiency, lift-to-drag ratio

1 Introduction

1.1 Motivation

The quest for aerodynamically efficient aircraft configurations has driven aerospace engineering for over a century. Many published optimization studies suffer from a subtle but significant issue: *constraint-solution inconsistency*. Authors declare constraints (e.g., Oswald efficiency $e > 0.5$) but report optimal solutions that violate these constraints (e.g., $e = 0.46$).

1.2 Contributions

1. **Self-consistent optimization framework**: Two constraint modes with guaranteed solution consistency.
2. **Transparent model hierarchy**: Clear distinction between first-principles, semi-empirical, and heuristic components.
3. **Physical realism**: Reynolds-number-dependent friction and structural penalties.
4. **Complete reproducibility**: All equations and algorithms provided.

2 Theoretical Framework

2.1 First-Principles Aerodynamics

The fundamental aerodynamic force equations:

$$L = \frac{1}{2}\rho V^2 S C_L, \quad D = \frac{1}{2}\rho V^2 S C_D \tag{1}$$

From Prandtl's lifting-line theory, the drag coefficient decomposes as:

$$C_D = C_{D0} + \frac{C_L^2}{\pi e \cdot AR} \tag{2}$$

2.2 Maximum Lift-to-Drag Ratio

Defining $K = 1/(\pi e \cdot AR)$, the lift-to-drag ratio is:

$$\frac{L}{D} = \frac{C_L}{C_{D0} + KC_L^2} \tag{3}$$

Differentiating and setting to zero yields:

$$C_{L,opt} = \sqrt{\frac{C_{D0}}{K}} = \sqrt{C_{D0} \cdot \pi e \cdot AR} \tag{4}$$

The maximum lift-to-drag ratio:

$$\boxed{\left(\frac{L}{D}\right)_{max} = \frac{1}{2}\sqrt{\frac{\pi e \cdot AR}{C_{D0}}}} \tag{5}$$

3 Semi-Empirical Correlations

3.1 Oswald Span Efficiency Factor

From Raymer (1992):

$$e = 1.78(1 - 0.045 \cdot AR^{0.68}) - 0.64 \cdot \cos^{0.5}(\Lambda) \cdot (1 - 0.25|\lambda - 0.4|) \tag{6}$$

Applicability: $4 < AR < 40$, $M < 0.6$.

3.2 Skin Friction Coefficient

Following Schlichting (1979), turbulent flat-plate friction:

$$C_f = \frac{0.455}{(\log_{10} Re)^{2.58}} \tag{7}$$

Applicability: $Re > 5 \times 10^5$.

3.3 Form Factor

From Hoerner (1965):

$$FF = 1 + 1.2 \cdot (t/c) + 100 \cdot (t/c)^4 \tag{8}$$

4 Heuristic Surrogate Models

Important: The following models are heuristic approximations, not first-principles relationships.

4.1 Camber Drag Bucket Model

$$\Delta C_{D0,camber} = k_{bucket} \cdot (C_{L,opt} - C_{L,design})^2 \tag{9}$$

where $k_{bucket} \approx 0.005$ and $C_{L,design} \approx 10 \times (f/c)$.

4.2 Structural Penalty Model

$$\Delta C_{D0,struct} = 0.0005 \cdot \left[\left(\frac{AR}{10} \right)^{1.5} - 1 \right], \quad AR > 10 \tag{10}$$

5 Optimization Methodology

5.1 Design Variables

Variable	Symbol	Range	Step	Points
Aspect ratio	AR	5.0–25.0	0.1	201
Sweep angle	Λ	0–45 deg	1 deg	46
Taper ratio	λ	0.20–0.80	0.05	13
Thickness	t/c	6–15%	1%	10
Camber	f/c	0–8%	1%	9

Total: $201 \times 46 \times 13 \times 10 \times 9 = 10,817,820$ configurations.

5.2 Constraint Modes

Mode A (Strict): $e > 0.5$, $AR \leq 25$ (conventional transport)
Mode B (Relaxed): $e > 0.3$, $AR \leq 40$ (HALE UAV)

5.3 Flight Conditions

Altitude: 3,000 m, Velocity: 50 m/s, Wingspan: 30 m (reference).

6 Results

6.1 Optimal Configuration

Parameter	Mode A	Mode B
Constraints		
e constraint	> 0.5	> 0.3
AR constraint	≤ 25	≤ 40
Optimal Geometry		
Aspect ratio AR	18.3	18.3
Sweep angle Λ	0 deg	0 deg
Taper ratio λ	0.40	0.40
Thickness t/c	6%	6%
Camber f/c	6%	6%
Performance		
$(L/D)_{max}$	**27.74**	**27.74**
Oswald e	0.5618	0.5618
$C_{L,opt}$	0.582	0.582
C_{D0}	0.01049	0.01049
Verification		
$e > e_{min}$?	Yes	Yes

6.2 Drag Breakdown

Component	C_{D0}	Percentage
Skin friction	0.00722	68.8%
Structural penalty	0.00074	7.0%
Form drag	0.00053	5.0%
Fuselage	0.00200	19.1%
Camber penalty	0.00001	0.1%
Total	**0.01049**	100%

6.3 Numerical Verification

$$K = \frac{1}{\pi \times 0.5618 \times 18.3} = 0.03096 \tag{11}$$

$$C_{L,opt} = \sqrt{\frac{0.01049}{0.03096}} = 0.582 \tag{12}$$

$$\left(\frac{L}{D}\right)_{max} = \frac{0.582}{2 \times 0.01049} = 27.74 \quad \checkmark \tag{13}$$

7 Discussion

7.1 Comparison with Literature

Aircraft	L/D	AR	Sweep
This work	27.7	18.3	0 deg
Sailplane	$\sim$60	$\sim$40	0 deg
Albatross	$\sim$20	$\sim$18	0 deg
Boeing 787	$\sim$20	$\sim$9	32 deg

7.2 Limitations

1. Low-fidelity aerodynamics (parabolic drag polar approximation)
2. Heuristic surrogates for camber and structure
3. Single-point optimization
4. No aeroelastic effects

8 Conclusion

We presented an exhaustive search framework for subsonic aircraft optimization with:

1. Self-consistent constraint handling
2. Transparent model hierarchy
3. Reynolds-dependent friction and structural penalties
4. Complete reproducibility

The optimal configuration achieves $(L/D)_{max} = 27.74$ at $AR = 18.3$ with zero sweep, 6% thickness, and 6% camber.

References

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